

Smart Door Lock System Using Arduino

Objective

Design a smart door lock system that allows access only when the correct password is entered using a keypad. If the password is correct, the green led turn on and unlock the door. Otherwise, access is denied.

Arduino Code

```
#include <Keypad.h>

const byte ROWS = 4;
const byte COLS = 3;

char keys[ROWS][COLS] =
{
  {'1','2','3'},
  {'4','5','6'},
  {'7','8','9'},
  {'*','0','#'}
};

byte rowPins[ROWS] = {0,1,2,3};
byte colPins[COLS] = {4,5,6};

Keypad keypad = Keypad(makeKeymap(keys),
                        rowPins,
                        colPins,
                        ROWS,
                        COLS);

String password = "1234";
String enteredPassword = "";

int greenLED = 12;
int redLED = 11;

void setup()
{
  pinMode(greenLED, OUTPUT);
  pinMode(redLED, OUTPUT);
}
```

```
}

void loop()
{
    char key = keypad.getKey();

    if(key)
    {
        if(key == '#')
        {
            if(enteredPassword == password)
            {
                digitalWrite(greenLED, HIGH);
                digitalWrite(redLED, LOW);

                delay(3000);

                digitalWrite(greenLED, LOW);
            }
            else
            {
                digitalWrite(redLED, HIGH);
                delay(2000);
                digitalWrite(redLED, LOW);
            }

            enteredPassword = "";
        }

        else if(key == '*')
        {
            enteredPassword = "";
        }

        else
        {
            enteredPassword += key;
        }
    }
}
```

How the System Works

Step 1

The user enters a password using the keypad.

Step 2

Press # to submit the password.

Step 3

The Arduino compares the entered password with the stored password (1234).

Step 4

If the password is correct:

- Green LED turns ON.
- Door unlocks for 3 seconds.

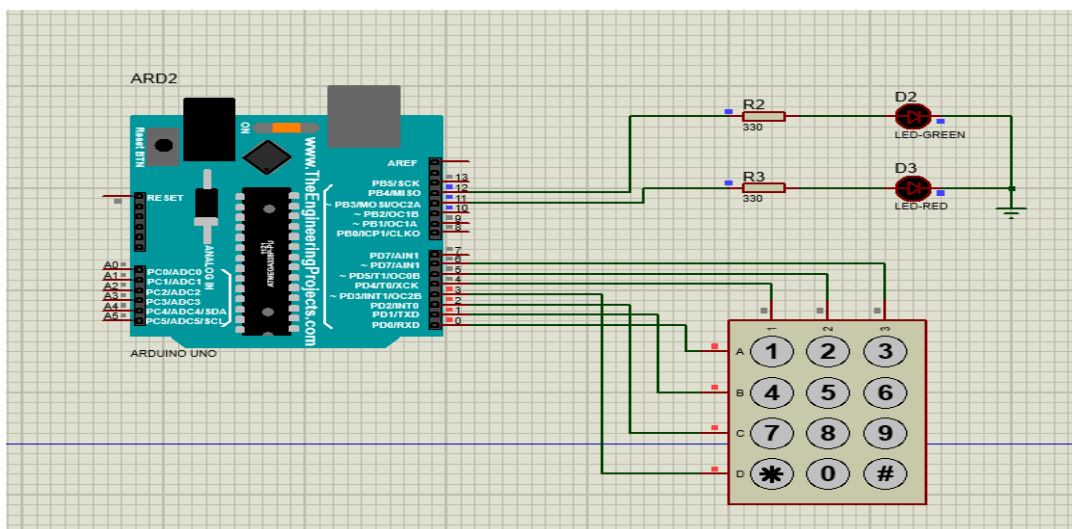
Step 5

If the password is incorrect:

- Red LED turns ON.
- Access is denied.

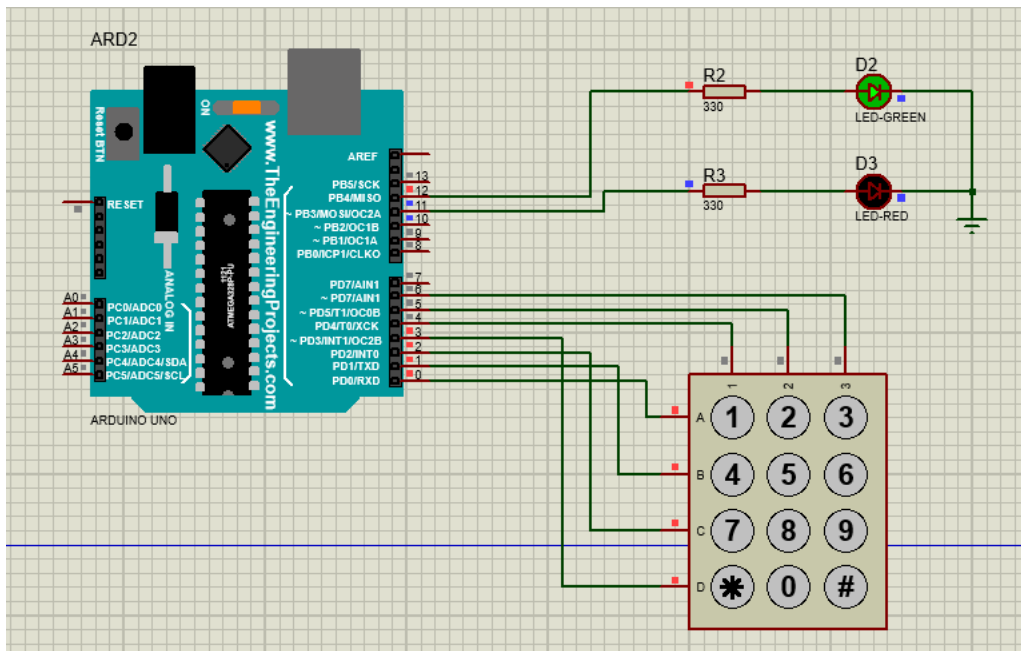
Output Images

Figure 1



Complete Proteus circuit.

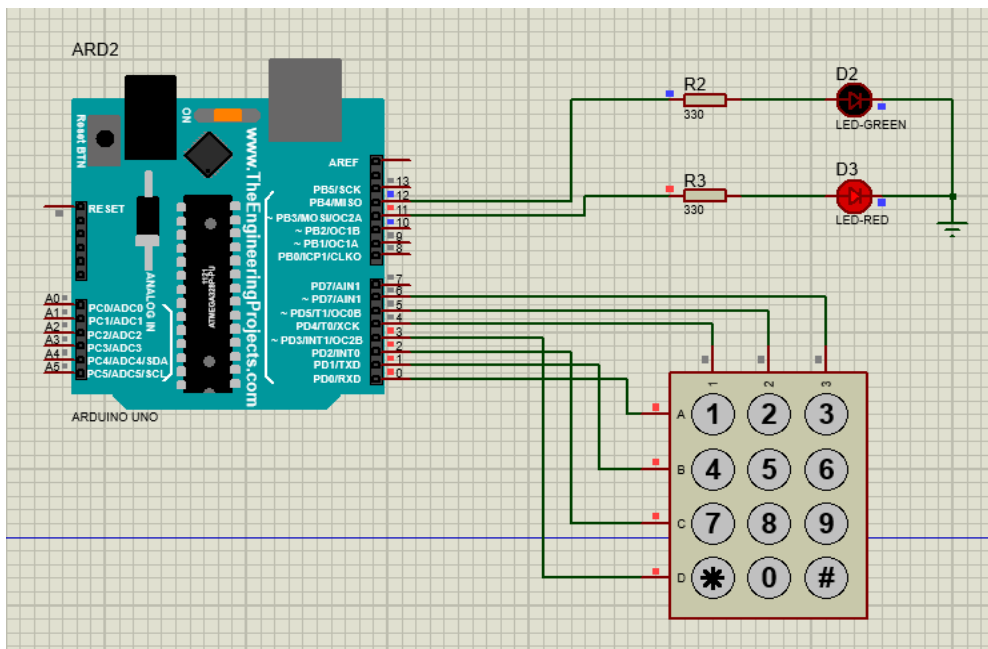
Figure 2



Correct Password (1234)

- Green LED ON
- Door Unlocked

Figure 3



Wrong Password

- Red LED ON
- Door Locked

Conclusion

The Smart Door Lock System enhances security by allowing access only to authorized users. The keypad is used for password entry, while the servo motor acts as the door lock mechanism. This project demonstrates a practical IoT security application that can be extended with RFID, fingerprint sensors, or Wi-Fi connectivity for remote access control.